

Errata of *Blocks of finite groups and their invariants*

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- p.5: The statement “Moreover, if $d \geq 3$ and $k_{d-2} \neq 0$, then the defect groups of B have maximal class.” appears as Lemma 4.7 in [237]. However, it seems that this result relies on OWC (see introduction of [237]).
- p.10: (3)(b) must be replaced by the following:

Every isomorphism $\varphi : R \rightarrow T$ in $\mathcal{F}$ can be extended to

$$N_\varphi := \{y \in N_P(R) : \exists z \in N_P(T) : \varphi(x^y) = \varphi(x)^z \ \forall x \in R\}.$$

(Thanks to Markus Linckelmann)

- p.10: subpairs are defined as (Q, b_Q) where b_Q is a block of $C_G(Q)$. Some people consider b_Q as a block of $Q C_G(Q)$ with the same idempotent. This is no big difference since there is only one block of $Q C_G(Q)$ covering b_Q . However, in order to define $(Q, b_Q) \trianglelefteq (P, b_P)$ via the Brauer correspondence $b_Q^{P C_G(Q)}$ one needs blocks of $Q C_G(Q)$ (otherwise the Brauer correspondence is not defined).
- The first sum in Conjecture 2.10 should start from $i = 0$.
- The relations $\sim_D$ in Proposition 5.1 should be replaced by $\sim_{\langle u \rangle}^D$ meaning that the corresponding elements are conjugate in $N_D(\langle u \rangle)$ (reported by René Reichenbach).
- In Theorem 6.4(2b), the case $(p, n) = (2, 1)$ must be excluded.
- In Theorem 6.9(vii), $p = 5$ is also possible. In fact, the case where N is a Sylow 5-subgroup of $\text{Aut}(\text{Sz}(32))$ was missed in the proof.
- In Lemma 6.13(i) and (ii) the group $C_{15} \times C_4$ should be replaced by $C_{15} \rtimes C_4$.
- At the end of the proof of Lemma 7.5, the claims about normalizers and centralizers are not correct. The argument is still elementary and left to the reader.
- The Cartan matrices in Theorem 8.1(5a) and (5b) should be interchanged.
- Second matrix on p. 189 should be

$$\begin{pmatrix} 4 & 1 & 2 & 2 \\ 1 & 4 & 2 & 2 \\ 2 & 2 & 4 & 1 \\ 2 & 2 & 1 & 4 \end{pmatrix}.$$

- The entry on position $(2, 1)$ of the matrix on p. 192 should be -1 instead of 1 .
- In the table 15.1 on p. 220 one can add the primitive group id 84 for the unique group of degree 59^2 (this information was not available at the time of writing).

An updated pdf version of the book can be downloaded from my homepage.

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